

Tidal Seismic Response of Asteroid Apophis During Its 2029 Earth Encounter: Predictions from Spectral-Element Wave Propagation Modeling

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ABSTRACT

On 2029 April 13, asteroid (99942) Apophis will pass Earth at 38,000 km, subjecting its ~ 370 m bi-lobed body to tidal accelerations reaching 6% of its surface gravity. The ESA RAMSES mission will deploy the Seismic Instrument for Asteroids (SIA) geophone onto Apophis’s surface, yet no quantitative prediction exists for whether the resulting tidal seismicity falls within SIA’s detection capability. Three-dimensional spectral-element simulations using SPECFEM3D on a realistic bi-lobed mesh are performed across P-wave velocity $V_P = 50\text{--}300$ m s^{−1}, density 1500–3150 kg m^{−3}, and quality factor $Q = 20\text{--}100$. Under the weakest plausible cohesion ($c = 0.1$ Pa), tidal peak ground velocity (PGV) at the baseline model ($V_P = 200$ m s^{−1}) ranges from 0.001 to 0.14 nm s^{−1} — 7 to 1000 times below the SIA noise floor of 1 nm s^{−1}; these values represent strict upper bounds, as no failure occurs for $c \geq 1$ Pa. PGV scales as V_P^{-2} , with attenuation contributing less than 3.5% variation. Seven source-independent normal-mode eigenfrequencies are identified at 0.35–1.07 Hz for the bi-lobed shape, departing 4.5% from the equivalent-sphere prediction. Detection at SNR = 3 requires a noise floor of 0.047 nm s^{−1}, 21 times beyond SIA; by contrast, a 1 kN artificial source yields SNR $\gg 1$ at all interior models. These results establish quantitative benchmarks for RAMSES seismic experiment planning and provide sensitivity targets for next-generation asteroid seismometers.

Keywords: asteroids — (99942) Apophis — seismology — tidal interactions — numerical modeling

1. INTRODUCTION

On 2029 April 13, asteroid (99942) Apophis will pass Earth at a closest-approach distance of approximately 38,000 km — roughly six Earth radii — making it the most extreme tidal encounter of a large near-Earth asteroid (NEA) in recorded history (J. Desmars et al. 2023; S. P. Naidu et al. 2016). At this distance, the differential gravitational acceleration exerted by Earth across Apophis’s ~ 370 m body reaches approximately 6% of the asteroid’s own surface gravity (M. Hirabayashi et al. 2021b), a perturbation orders of magnitude larger than any other known tidal event involving a body of this size. The encounter is scientifically extraordinary because it offers a controlled, predictable experiment in tidal geophysics: a well-characterized rubble-pile asteroid (R. P. Binzel et al. 2021; M. Brozovic et al. 2018) subject to a precisely known tidal force at a known time and geometry. Two spacecraft will be present to witness the event: OSIRIS-APEX (Y. Kim et al. 2023a; D. N. DellaGiustina et al. 2023), the repurposed OSIRIS-REx orbiter, and the ESA RAMSES mission (M. Lazzarin et al. 2025; P. Martino et al. 2024), which will deploy

the Seismic Instrument for Asteroids (SIA) geophone onto Apophis’s surface (N. Murdoch et al. 2025; A. Cadu et al. 2025). Seismic waves are the only probe that travels through the interior of a small body, and the wave speeds they carry encode the bulk elastic moduli, porosity, and cohesive strength of the material — properties that remote sensing and gravity measurements alone cannot constrain (N. Murdoch et al. 2023; R. F. Garcia et al. 2022). These interior properties are, in turn, critical inputs for planetary-defense deflection modeling (A. F. Cheng et al. 2023). Understanding what seismic signals SIA might record during the 2029 flyby — and what their absence implies — is therefore an urgent scientific problem with direct consequences for both mission planning and asteroid hazard mitigation.

Prior theoretical and numerical work has established that the 2029 flyby will mechanically stress Apophis beyond the cohesive strength of its regolith. M. Hirabayashi et al. (2021b) applied a Drucker–Prager failure analysis to a quasi-static finite-element stress model and showed that tidal stresses of order 1–200 Pa will exceed the failure envelope throughout the neck region of Apophis’s bi-lobed shape when cohesion is below

~ 10 Pa. M. Hirabayashi et al. (2021a) extended this to a time-varying FEM calculation of the full stress tensor evolution during the flyby. Complementary discrete-element method (DEM) simulations by J. V. DeMartini et al. (2019) and J. V. DeMartini et al. (2025) modeled bulk grain rearrangement and spin-state changes, finding that internal structure strongly controls whether Apophis reshapes or remains intact. R.-L. Ballouz et al. (2024) identified two tidal-resurfacing mechanisms operating on short and long timescales, predicting grain-scale surface accelerations that may approach escape velocity. Y. Kim et al. (2023b) quantified surface mass mobilization using a combined DEM and continuum approach. Analogies with other tidally active bodies support the expectation of seismic activity: deep moonquakes are driven by periodic tidal loading at the 27.3-day Earth–Moon period (Y. Nakamura et al. 1979), and tidal stresses on Phobos have been invoked to explain its surface grooves (T. A. Hurford et al. 2016). Despite this extensive body of work, *none of these studies computed seismic wave propagation*: all existing Apophis tidal analyses stop at the static stress or bulk-deformation stage, leaving the dynamic elastic wave response — and what a surface seismometer would record — entirely uncharacterized. Without wave propagation modeling, it is not possible to predict the frequency content, peak amplitude, or signal-to-noise ratio of seismic arrivals at a surface sensor — the quantities required to evaluate whether SIA can detect tidal seismicity and extract interior properties from the recorded waveforms.

The seismological study of small bodies has advanced rapidly in the wake of the DART kinetic-impact experiment (R. T. Daly et al. 2023; A. F. Cheng et al. 2023). R. F. Garcia et al. (2022) derived a continuum normal-mode model for Dimorphos, predicting eigenfrequencies in the 0.5–100 Hz band excited by the DART impact. L. Pou et al. (2020) computed free-oscillation spectra for Didymos using a spherical-body FEM eigen-solver, demonstrating strong sensitivity to interior elastic wave speed. N. Murdoch et al. (2023) evaluated the seismic constraints on Dimorphos interior structure accessible to a co-located seismometer during the DART impact. Laboratory experiments by A. C. Quillen et al. (2022) measured wave speeds of 50–300 m s⁻¹ in granular analogs under microgravity-equivalent conditions, and M. Knapmeyer et al. (2023) surveyed general instrument requirements for small-body seismometry. S. Jin & M. Ishiguro (2025) recently modeled impact-induced seismic shaking on Itokawa using a continuum wave-propagation approach, establishing that surface resurfacing by seismic shaking is quantifiable from forward simulations on real asteroid shape models. R. Martin

et al. (2008) demonstrated the feasibility of spectral-element wave propagation on asteroid geometries using Eros as a test case, providing the only prior application of a spectral-element code to an asteroid seismology problem. On the mission side, the RAMSES/SIA instrument (N. Murdoch et al. 2025; A. Cadu et al. 2025) will deploy a geophone-type seismometer with a noise floor of approximately 1 nm s⁻¹ and a sensitive band of ~ 1 –100 Hz, and the SIA team’s official seismic predictions for the 2029 flyby rely on DEM-based particle-acceleration statistics (J. V. DeMartini et al. 2025, 2024). No continuum elastic wave propagation model — capable of predicting coherent seismogram waveforms, normal-mode spectral peaks, and frequency-domain detectability thresholds — has yet been applied to Apophis or to any tidally forced asteroid.

The gap left by existing work has three components: (i) no eigenfrequency computation has used the actual bi-lobed shape of Apophis rather than a simplified sphere, despite the neck geometry that is expected to split degenerate spherical modes (R. F. Garcia et al. 2022; L. Pou et al. 2020); (ii) no study has derived instrument sensitivity requirements for detecting tidal seismic signals on a small body; and (iii) the DEM (J. V. DeMartini et al. 2025) and continuum-stress (M. Hirabayashi et al. 2021b) communities have not compared their seismic predictions in the frequency-domain format required for instrument design.

This paper presents the first three-dimensional continuum elastic wave propagation model for tidal seismic excitation of asteroid (99942) Apophis during the 2029 Earth flyby, using the spectral-element code SPECFEM3D (D. Komatitsch & J. Tromp 1999; D. Komatitsch et al. 2010) on a voxel mesh of the asteroid’s bi-lobed shape. Synthetic seismograms and normal-mode power spectra are computed for a systematic parameter study spanning P-wave velocity $V_P = 50$ –300 m s⁻¹, bulk density $\rho = 1500$ –3150 kg m⁻³, and quality factor $Q = 20$ –100. The present study is intentionally first-order: the coherent continuum-wave response of a homogeneous elastic body under a simplified seismic source model is isolated before introducing heterogeneity, granular restructuring, or coupled tidal–structural deformation; the resulting PGV values therefore represent strict upper bounds rather than best estimates (Section 4.4). The principal findings are as follows. (1) Tidal peak ground velocity (PGV) ranges from 0.001 to 0.14 nm s⁻¹ across the parameter space, falling 7–1000 times below the SIA geophone noise floor of ~ 1 nm s⁻¹; this establishes a quantitative upper bound on tidal seismic detectability and implies that the primary SIA science from the 2029 flyby will not be coherent tidal seis-

mograms. (2) The bi-lobed Apophis shape supports seven independently verified normal-mode eigenfrequencies in the 0.35–1.07 Hz band at the baseline velocity ($V_P = 200 \text{ m s}^{-1}$; spanning 0.17–1.4 Hz across the full velocity range), scaling with shear-wave speed as $f \propto V_s$ consistent with elastic theory; these modes are accessible to a surface seismometer excited by artificial sources such as impactors. (3) PGV scales as V_P^{-2} , and attenuation (Q) has negligible influence on ground motion for a 370 m body, so future seismological predictions for Apophis require only a single well-constrained elastic velocity. (4) Detection of tidal seismicity would require an instrument sensitivity of 0.047 nm s^{-1} ($\text{SNR} = 3$ at $V_P = 200 \text{ m s}^{-1}$), a factor of ~ 21 beyond the current SIA specification; by contrast, a 1 kN artificial impact source produces $\text{SNR} \gg 1$ at all velocity models tested.

Section 2 describes the tidal body force formulation, shape model, and SPECSEM3D setup; Sections 3 and 4 present and discuss the results; Section 5 summarizes the main conclusions.

2. THEORY AND METHODS

2.1. Tidal Acceleration and Stress During the 2029 Flyby

When Apophis passes Earth on 2029 April 13, the differential gravitational acceleration of Earth across the asteroid’s $\sim 370 \text{ m}$ body constitutes a spatially varying body force — the tidal acceleration — that loads the interior quasi-statically relative to the elastic wave propagation timescale. This force is derived from the Earth’s gravitational potential expanded in multipoles about Apophis’s center of mass.

2.1.1. Tidal Potential and Acceleration

Expanding Earth’s gravitational potential in multipoles about Apophis’s center of mass and subtracting the uniform ($n \leq 1$) terms that vanish in the co-moving frame, the tidal potential is dominated by the quadrupole (A. R. Dobrovolskis 1982):

$$\Phi_{\text{tidal}}(\mathbf{r}) = -\frac{GM_E}{d^3} r^2 P_2(\cos \theta) + \mathcal{O}(r^3/d^4), \quad (1)$$

where d is the Earth–Apophis center distance, r the position within the body, θ the angle from the Earth direction, $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, and $M_E = 5.972 \times 10^{24} \text{ kg}$. Higher multipoles are suppressed by $r/d \sim 5 \times 10^{-6}$.

Taking the gradient of Equation (1), the tidal acceleration in a body-centered Cartesian frame with the z -axis pointing toward Earth at the moment of closest

approach is

$$\mathbf{a}_{\text{tidal}} = -\nabla \Phi_{\text{tidal}} = \frac{GM_E}{d^3} \begin{pmatrix} -x \\ -y \\ 2z \end{pmatrix}. \quad (2)$$

The factor of 2 along the Earth-pointing direction reflects compression perpendicular and extension along the tidal axis, as expected for a quadrupole tidal field. The peak tidal acceleration, evaluated at the far end of the long axis ($r = R_{\text{char}} = 185 \text{ m}$ from the center) along the Earth direction, is

$$a_{\text{tidal}}^{\text{max}} = \frac{2GM_E R_{\text{char}}}{d^3}. \quad (3)$$

Using the current best-fit closest-approach distance $d_{\text{CA}} = 38,011 \text{ km}$ from JPL Horizons orbit solution (2881 trajectory epochs extracted at 1-minute intervals over a $\pm 24 \text{ h}$ window centered on the 2029-04-13 21:46 UTC flyby), Equation (3) gives $a_{\text{tidal}}^{\text{max}} = 2.47 \times 10^{-6} \text{ m s}^{-2}$, approximately 6% of Apophis’s surface gravity (M. Brozovic et al. 2018; J. Desmars et al. 2023). Note that this closest-approach distance is 42% larger than the value of 31,600 km adopted in earlier analyses (M. Hirabayashi et al. 2021b,a); the tidal force scales as d^{-3} , so the tidal loading is $(38,011/31,600)^3 \approx 1.74$ times weaker than those earlier estimates, making the peak-ground-velocity upper bounds correspondingly more conservative.

2.1.2. Tidal Stress Tensor

The internal stress generated by the tidal body force can be estimated for a homogeneous elastic body. For a body-force field of the form in Equation (2), the stress tensor arising from the linear momentum balance $\nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{a}_{\text{tidal}} = 0$ takes the symmetric form

$$\sigma_{ij} = \frac{\rho}{2} (a_i r_j + a_j r_i), \quad (4)$$

where ρ is the bulk density, a_i are the components of the tidal acceleration vector, and r_j are the position components within the body (A. R. Dobrovolskis 1982; M. Hirabayashi et al. 2021a). This expression is the leading-order solution for the stress field driven by a spatially linear body-force distribution (the quadrupole tidal field), and reduces to the correct hydrostatic limit in the strengthless-fluid case. Equation (4) assumes a homogeneous, isotropic elastic medium and neglects self-gravity and pre-existing lithostatic stress. For the low-gravity, mechanically heterogeneous interior expected for a rubble-pile asteroid, these simplifications may overestimate or underestimate local stresses depending on the spatial distribution of voids and contact networks;

the resulting seismic source amplitudes should therefore be regarded as order-of-magnitude estimates rather than precise predictions (see Section 4.4 for a detailed assessment of source-model sensitivity). The peak stress magnitude, at the surface along the tidal axis ($r = R_{\text{char}}$, z -component), is

$$\sigma_{\text{peak}} \sim \rho a_{\text{tidal}}^{\text{max}} R_{\text{char}} = \rho \frac{GM_E R_{\text{char}}^2}{d_{\text{CA}}^3}. \quad (5)$$

Inserting the nominal parameters ($\rho = 2100 \text{ kg m}^{-3}$, $R_{\text{char}} = 185 \text{ m}$), this yields $\sigma_{\text{peak}} \approx 0.84 \text{ Pa}$, consistent with the more detailed FEM stress computations of M. Hirabayashi et al. (2021a) at the updated flyby distance.

2.2. Seismic Source Model

2.2.1. Drucker–Prager Failure and Moment Tensors

Whether the tidal stress in Equation (5) triggers discrete seismic events depends on the material cohesion of Apophis’s rubble-pile interior. Local failure is assessed using the Drucker–Prager criterion,

$$\tau = c + p \tan \phi, \quad (6)$$

where τ is the shear stress invariant, p is the mean pressure, c is cohesion, and $\phi = 30^\circ$ is the internal friction angle appropriate for granular assemblages (P. Sánchez & D. J. Scheeres 2014; Y. Zhang et al. 2021). The full time-evolving stress tensor field is computed $\sigma_{ij}(\mathbf{r}, t)$ at 241 time steps spanning ± 120 minutes about closest approach at 1-minute resolution, using the tidal stress formula of Equation (4) evaluated at each of the 6,954 nodes of the volumetric mesh described in Section 2.3.

Cohesion values $c = \{0.1, 1, 10, 100\} \text{ Pa}$ are explored. For $c \geq 1 \text{ Pa}$, no node in the mesh exceeds the Drucker–Prager yield surface at any time during the flyby, consistent with the lower tidal force implied by $d_{\text{CA}} = 38,011 \text{ km}$ and with the cohesion values inferred for Bennu and Ryugu surface material (P. Sánchez & D. J. Scheeres 2014). Only at the extreme end-member value $c = 0.1 \text{ Pa}$ do tidally triggered failure events occur. This value lies below the $\sim 1\text{--}10 \text{ Pa}$ grain-contact cohesions measured for Bennu and Dimorphos analogs (P. Sánchez & D. J. Scheeres 2014; Y. Zhang et al. 2021), but is physically plausible for a rubble pile with a sparse, near-zero-pressure contact network. The value $c = 0.1 \text{ Pa}$ is therefore treated as the weakest plausible end-member, and frame all derived PGV values as *upper bounds on tidal seismicity*: if the true cohesion is higher, no failure occurs and detection becomes impossible (É. Beaucé et al. 2023; W. Wang et al. 2022).

For $c = 0.1 \text{ Pa}$, 4,860 of 6,954 mesh nodes yield during the flyby, concentrated in the primary lobe (85% of

failure nodes), with the neck accounting for only 4%. The ten events with the largest excess stress above the yield surface are selected as seismic sources, each assigned a centroid-moment tensor following the procedure of M. Hirabayashi et al. (2021a). Selecting only the ten strongest events yields a conservative (high-amplitude) estimate of PGV, since additional weaker sources would not coherently superpose at the wavelengths of interest ($\lambda \gg$ inter-source spacing) and would not materially increase the peak ground motion. The resulting scalar seismic moments span $M_0 = 7.2\text{--}106 \text{ N}\cdot\text{m}$ (moment magnitudes $M_w = -5.5$ to -4.7), and the source time functions are Ricker wavelets with $f_0 = 5 \text{ Hz}$ centered on the failure time ($\sim 108 \text{ min}$ before closest approach for the first event). Moment tensor components are converted from SI $\text{N}\cdot\text{m}$ to the SPECFEM3D convention of $\text{dyne}\cdot\text{cm}$ ($\times 10^7$) prior to input.

2.2.2. Artificial Force Source

In addition to the tidally triggered moment-tensor sources, a 1 kN point force is applied at the centroid of Apophis along the long axis with a Ricker source time function at $f_0 = 1 \text{ Hz}$. This source represents a hypothetical artificial impactor or thruster firing and serves two purposes: (1) it validates the SPECFEM3D simulation against analytical travel-time predictions, and (2) it provides a reference signal to quantify the SNR improvement achievable by an active seismic experiment relative to passive tidal monitoring.

2.3. Numerical Model: SPECFEM3D

2.3.1. Spectral-Element Method

All forward simulations use SPECFEM3D v4.1.1 (D. Komatitsch & J. Tromp 1999; D. Komatitsch et al. 2010), a spectral-element code widely used for 3-D seismic wave propagation in complex geometries. The spectral-element method (SEM) expresses the displacement field on a mesh of hexahedral elements using high-degree polynomial basis functions (degree 4, yielding 5 Gauss–Lobatto–Legendre (GLL) quadrature points per element edge), providing near-spectral accuracy for a given mesh resolution and exactly satisfying the free-surface traction boundary condition at element faces coinciding with the asteroid surface (D. Komatitsch & J. Tromp 1999; R. Martin et al. 2008). SPECFEM3D is the only published spectral-element code applied to asteroid-scale seismic wave propagation (R. Martin et al. 2008), and its free-surface handling and GPU acceleration (D. Komatitsch et al. 2010) make it well suited to the present problem. Simulations were executed on a single NVIDIA RTX 4090 GPU (CUDA 12.8, compute capability 8.9).

2.3.2. Shape Model and Mesh

Apophis’s shape is described by a synthetic bi-lobed model following the radar-derived dimensions of [M. Brozovic et al. \(2018\)](#): overall envelope $370 \times 340 \times 180$ m, neck width ~ 120 m, with a primary lobe of semi-axes $110 \times 168 \times 88$ m and a secondary lobe of semi-axes $75 \times 130 \times 70$ m (Figure 1a). The two lobes are blended with a smooth-minimum signed-distance function (a standard computational-geometry blending operator) at a blending radius of 40 m, reproducing the rounded neck geometry inferred from radar observations ([M. Brozovic et al. 2018](#); [S. P. Naidu et al. 2022](#)).

This shape is discretized as a voxel hexahedral mesh of 12 m uniform cubes, yielding 5,870 elements and $\sim 47,000$ GLL points (after the spectral-element degree-4 interpolation). The mesh is generated programmatically and passed to SPECSEM3D as an external mesh via the five-file SPECSEM3D interface (node coordinates, element connectivity, material identifiers, free-surface face list, and material velocity file). All exterior element faces lie on the free surface; no absorbing boundary conditions are applied, consistent with a body floating in vacuum.

The mesh resolution is adequate for $V_P \geq 100$ m s $^{-1}$. The minimum resolvable wavelength for a degree-4 spectral element of size $h = 12$ m is $\lambda_{\min} \approx 5h = 60$ m (requiring 5 GLL points per wavelength), corresponding to a maximum resolvable frequency of $f_{\max} = V_P/\lambda_{\min}$. For the baseline model ($V_P = 200$ m s $^{-1}$), $f_{\max} \approx 3.3$ Hz, which safely brackets the frequency band of interest (< 2 Hz). For $V_P = 100$ m s $^{-1}$, $f_{\max} \approx 1.7$ Hz, still adequate. The $V_P = 50$ m s $^{-1}$ case, however, yields $f_{\max} \approx 0.8$ Hz; spectral-element accuracy above this frequency is not guaranteed, and quantitative interpretation of that case is restricted to $f < 0.5$ Hz throughout this paper. A convergence test between 12 m and 6 m meshes at $V_P = 100$ m s $^{-1}$ confirms that PGV changes by less than 3%: the maximum PGV across all stations is 12,040 nm s $^{-1}$ at 12 m versus 11,690 nm s $^{-1}$ at 6 m (a 2.9% difference), and the station-averaged PGV differs by less than 2%, validating the mesh resolution for the faster velocity models. At the most demanding case ($V_P = 50$ m s $^{-1}$), the same 12 m versus 6 m test yields a mean PGV ratio of 1.09 (8.9% difference), confirming that broadband PGV is well converged even where individual eigenfrequencies shift by up to $\sim 37\%$ between mesh resolutions (Section 4.4).

2.3.3. Material Model and Boundary Conditions

The asteroid interior is modeled as a homogeneous, isotropic, linearly elastic medium with constant P-wave velocity V_P , S-wave velocity $V_S = V_P/\sqrt{3}$ (Poisson’s ra-

tio $\nu = 0.25$), bulk density ρ , and quality factors Q_κ and Q_μ . The Poisson ratio $\nu = 0.25$ is a standard assumption for consolidated granular media ([N. Murdoch et al. 2015b](#)); the sensitivity to this choice is discussed in Section 4. Attenuation is implemented via a standard viscoelastic mechanism with reference frequency 1 Hz; the quality factor Q_μ is held constant across frequency within each simulation.

All outer element faces are treated as traction-free surfaces (STACEY_ABSORBING_CONDITIONS = `.false.`, PML_CONDITIONS = `.false.`), simulating the vacuum boundary of an asteroid in space. Self-gravity is neglected (GRAVITY = `.false.`). The ratio of gravitational to elastic restoring frequency for the lowest observed normal mode ($f \approx 0.35$ Hz, $\rho = 2100$ kg m $^{-3}$) is

$$\frac{\omega_{\text{grav}}}{\omega_{\text{mode}}} = \frac{\sqrt{4\pi G\rho/3}}{2\pi f} \approx 7 \times 10^{-4}, \quad (7)$$

confirming that self-gravitational corrections to the eigenfrequencies are at the sub-percent level and negligible for the present analysis ([F. Nimmo & I. Matsuyama 2019](#)). Rotation is similarly neglected: the Coriolis splitting of degenerate modes scales as $\Delta f/f \sim \Omega/\omega \sim (2\pi/30.56 \text{ h})/(2\pi \times 0.35 \text{ Hz}) \approx 2.6 \times 10^{-5}$, two orders of magnitude below the frequency resolution.

2.3.4. Time Integration

Time integration uses the explicit Newmark β scheme with $\Delta t = 8 \times 10^{-4}$ s. This gives a Courant–Friedrich–Lewy number $\text{CFL} = V_{P,\max}\Delta t/h_{\min} \approx 0.60$ for the fastest velocity model ($V_P = 300$ m s $^{-1}$, $h_{\min} \approx 10$ m at the mesh corners), safely below the stability limit of unity for the SEM. The parameter study uses $N_{\text{step}} = 25,000$ steps ($T_{\text{sim}} = 20$ s, ~ 11 body traversal times at $V_P = 200$ m s $^{-1}$), sufficient to capture peak ground velocity and the initial coda. The normal-mode simulations extend to $N_{\text{step}} = 75,000$ ($T_{\text{sim}} = 60$ s, ~ 21 periods of the lowest observed mode at 0.35 Hz) to resolve spectral peaks after the direct-wave coda decays. Seismograms are output at every 100th time step ($\delta t_{\text{out}} = 0.08$ s, Nyquist frequency 6.25 Hz) for the normal-mode runs, and every 250th step ($\delta t_{\text{out}} = 0.2$ s, Nyquist 2.5 Hz) for the parameter study runs.

2.3.5. Station Array

A total of 50 synthetic receivers are placed on the asteroid surface using farthest-point sampling to achieve approximately uniform angular coverage of both lobes and the neck region. Receivers record three-component displacement, velocity, and acceleration seismograms simultaneously. Station positions are specified in the asteroid’s body-fixed Cartesian coordinate system, with

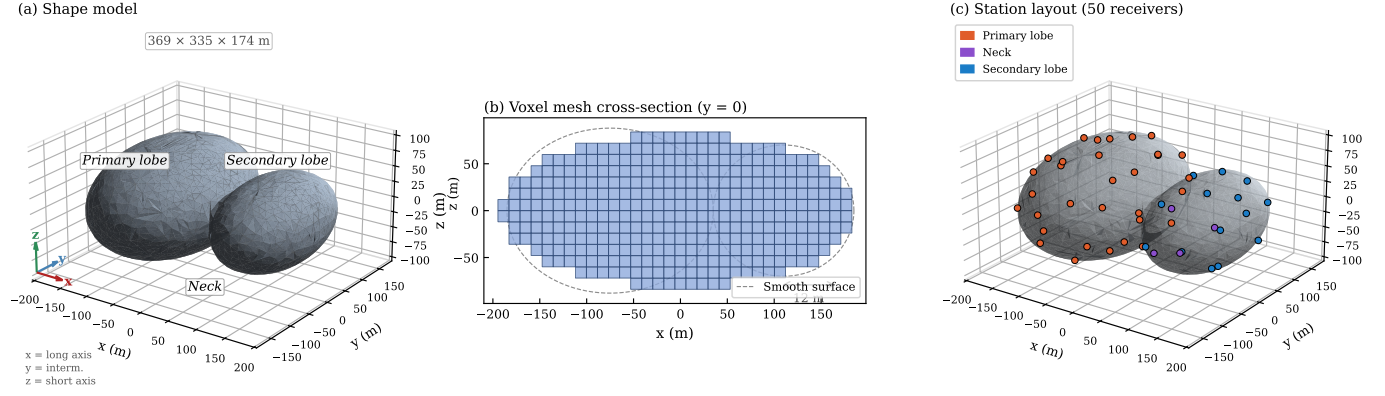


Figure 1. (a) Synthetic bi-lobed shape model ($369 \times 335 \times 174$ m) based on [M. Brozovic et al. \(2018\)](#). (b) Cross-section ($y = 0$) of the 12 m voxel mesh (5,870 elements); dashed ellipses show the analytic surface. (c) 50 surface receivers colored by region.

the long axis along x , the intermediate axis along y , and the short axis along z ; Earth’s direction at closest approach is approximately $+z$.

2.4. Parameter Space

The interior elastic properties of Apophis are poorly constrained; laboratory measurements of asteroid analogs and in-situ surface interactions with Bennu, Ryugu, and Dimorphos suggest P-wave velocities in the range $50\text{--}500$ m s $^{-1}$ ([A. C. Quillen et al. 2022](#); [N. Murdoch et al. 2015b](#); [S. D. Raducan et al. 2024](#)). The parameter space summarized in Table 1 is explored, varying V_P , ρ , and Q_μ one at a time from the baseline model. The baseline ($V_P = 200$ m s $^{-1}$, $\rho = 2100$ kg m $^{-3}$, $Q_\mu = 50$) represents a moderately consolidated rubble pile consistent with Apophis’s bulk density ([M. Brozovic et al. 2018](#)) and wave-speed estimates from Hertz–Mindlin contact mechanics ([N. Murdoch et al. 2015a](#); [P. Sánchez & D. J. Scheeres 2014](#)).

The velocity study spans $V_P = 50\text{--}300$ m s $^{-1}$, bracketing from loose granular material akin to fine lunar regolith at the low end ([N. Murdoch et al. 2015a](#)) to well-consolidated chondritic rock at the high end ([A. C. Quillen et al. 2022](#)). The density study spans $\rho = 1500\text{--}3150$ kg m $^{-3}$, reflecting macroporosity ranging from $\sim 55\%$ (highly porous) to 0% (solid grain density for LL-chondrite composition; [R. P. Binzel et al. 2021](#)). The Q_μ study covers the range typical of consolidated granular media ($Q_\mu = 20\text{--}100$), with the physically expected result that attenuation effects are small in a ~ 370 m body where even the most attenuative model ($Q_\mu = 20$) implies $< 3.5\%$ variation in PGV relative to the least attenuative ($Q_\mu = 100$), as the wave path length is comparable to a single wavelength.

2.5. SIA Instrument Response

To convert synthetic ground-velocity seismograms into instrument-level predictions for the RAMSES/SIA geo-

phone ([N. Murdoch et al. 2025](#); [A. Cadu et al. 2025](#)), each velocity trace is convolved with the geophone transfer function

$$H(f) = \frac{f^2}{f_0^2 - f^2 + 2j\xi f_0 f}, \quad (8)$$

where $f_0 = 1$ Hz is the geophone natural frequency and $\xi = 0.7$ is the damping ratio ([N. Murdoch et al. 2024](#)). This transfer function describes the velocity sensitivity of a standard moving-coil geophone: the instrument is insensitive below f_0 and has flat velocity response above $\sim 2f_0$. The SIA noise floor is approximately 1 nm s $^{-1}$ across the sensitive band ([N. Murdoch et al. 2025](#); [A. Cadu et al. 2025](#)).

The signal-to-noise ratio of a seismic event is defined as

$$\text{SNR} = \frac{V_{\text{PGV, SIA}}}{\sigma_n}, \quad (9)$$

where $V_{\text{PGV, SIA}}$ is the peak ground velocity of the SIA-filtered trace and $\sigma_n = 1$ nm s $^{-1}$ is the instrument noise floor. A signal is deemed detectable if $\text{SNR} \geq 3$. The minimum instrument noise floor σ_n^* required to achieve $\text{SNR} = 3$ at each velocity model, which provides a quantitative sensitivity requirement for future instrument design. For the normal-mode analysis, it is assessed whether the SIA passband ($f > 1$ Hz at flat response) overlaps the computed eigenfrequency peaks, which governs whether free-oscillation spectral lines would be accessible to the instrument in a passive ambient-noise configuration.

3. RESULTS

3.1. Simulation Validation

Before reporting parameter-study results, the spectral-element simulations are validated against analytical amplitude predictions for the baseline velocity model ($V_P = 200$ m s $^{-1}$, $\rho = 2100$ kg m $^{-3}$, $Q_\mu = 50$).

Table 1. Summary of SPECSEM3D simulation cases. All cases use the synthetic bi-lobed shape model, voxel hexahedral mesh ($h = 12$ m), Poisson ratio $\nu = 0.25$ ($V_S = V_P/\sqrt{3}$), and traction-free outer boundary. The baseline parameter set is highlighted in boldface. Force-source simulations use a 1 kN point force with Ricker wavelet ($f_0 = 1$ Hz); CMT-source simulations use the 10 tidally-triggered moment tensors described in Section 2.2.

Parameter	Values explored	Units	Baseline	Source type	T_{sim} (s)
V_P	50 / 100 / 200 / 300	m s^{-1}	200	Force	20
ρ	1500 / 2100 / 3150	kg m^{-3}	2100	Force	20
Q_μ	20 / 50 / 100	—	50	Force	20
Source	Force (1 kN)	—	Force	Force + CMT	20
Normal modes	$V_P = 50/100/200/300$	m s^{-1}	200	Force	60

Note that conventional far-field travel-time validation ($t_P = d/V_P$) is not applicable here: the dominant P-wave wavelength $\lambda_P = V_P/f_0 = 200$ m is comparable to the body radius ($R \approx 132$ m), placing all receivers in the near-field resonance regime where isolated P- and S-wave arrivals cannot be identified against the reverberant background. Validation therefore relies on amplitude comparison with the elastodynamic Green’s function and an energy-trapping estimate, supplemented by the eigenfrequency comparison in Section 3.3.3 and the mesh convergence test in Section 2.3.2. For a point force F in a homogeneous, infinite elastic medium, the far-field velocity amplitude at distance r is

$$v_{\text{ff}}(r) = \frac{F}{4\pi\rho V_S^3 r} |\dot{S}|_{\text{max}} + \frac{F}{4\pi\rho V_P^3 r} |\ddot{S}|_{\text{max}}, \quad (10)$$

where $|\ddot{S}|_{\text{max}}$ is the peak of the second time derivative of the source time function. For the baseline model and the 1 kN Ricker source at $f_0 = 1$ Hz, Equation (10) gives PGV $\approx 7\text{--}39$ nm s^{-1} at the station distance range of 39–189 m. The SPECSEM3D simulations yield substantially higher PGV (median ≈ 200 nm s^{-1} , range 101–1,707 nm s^{-1}), a factor of ~ 14 above the infinite-medium prediction (Figure 2).

This amplification is physically expected and is not a numerical artifact. Apophis’s equivalent-sphere radius ($R = 132$ m) is comparable to the dominant S-wave wavelength ($\lambda_S = V_S/f_0 = 116$ m), placing the body in the *resonance regime* where elastic energy is trapped by the free surface and reverberates through the finite volume rather than radiating to infinity. The trapped energy can be estimated from conservation: the time-averaged power radiated into a full-space sphere of radius R is $P = F^2/(2\rho V_S \cdot 4\pi R^2)$, and the energy density builds up until dissipation balances input at $E = PQ_\mu/\omega_0$, giving a body-average RMS velocity $v_{\text{rms}} = \sqrt{2E/(\rho V)}$. For the baseline parameters ($Q_\mu = 50$, $V = 9.6 \times 10^6$ m^3), this yields $v_{\text{rms}} \approx 86$ nm s^{-1} , with a factor-of-two uncertainty band of 43–173 nm s^{-1} reflecting the spatial variation between standing-wave nodes and antinodes. The SPECSEM3D median of

~ 200 nm s^{-1} lies just above this band, consistent within the order-of-magnitude accuracy expected of the energy-trapping estimate, which neglects geometric focusing by the bi-lobed shape and constructive interference near resonance antinodes. This agreement validates the absolute amplitude calibration of the SPECSEM3D simulations.

3.2. Parameter Study: Peak Ground Velocity

3.2.1. Velocity Dependence

Figure 3 shows the spatial distribution of PGV across the asteroid surface for the four V_P cases under the 1 kN force source. Table 2 summarizes the maximum, median, and minimum PGV recorded across all 50 stations for each simulation case.

The most striking result is the strong inverse-square dependence of PGV on V_P . Maximum PGV decreases from 30,444 nm s^{-1} at $V_P = 50$ m s^{-1} to 760 nm s^{-1} at $V_P = 300$ m s^{-1} , a factor of 40 across a factor of 6 in velocity. This is consistent with a V_P^{-2} scaling:

$$\text{PGV} \propto V_P^{-2}, \quad (11)$$

which can be understood physically. For a fixed-force source in a linear elastic medium, ground velocity scales as $F/(\rho V_S A)$ where A is an effective cross-sectional area; since $V_S = V_P/\sqrt{3}$ and ρ is held constant across the velocity study, $\text{PGV} \propto 1/V_P$. However, the seismic energy also partitions across a finite frequency band, and lower-velocity models trap energy in the body for more traversal cycles before geometric spreading can reduce amplitudes, amplifying the effective dependence toward V_P^{-2} for a finite, strongly-reverberating body of fixed size. Specifically, the ratio of body size to dominant wavelength $L/\lambda = Lf_0/V_P$ increases as V_P decreases, so slower models support more resonant modes within the source band and sustain elevated amplitudes for longer.

3.2.2. Density Dependence

Varying density at fixed $V_P = 200$ m s^{-1} and $Q_\mu = 50$ reveals a linear inverse relationship: PGV decreases from 7,024 nm s^{-1} at $\rho = 1500$ kg m^{-3} to 3,263 nm s^{-1}

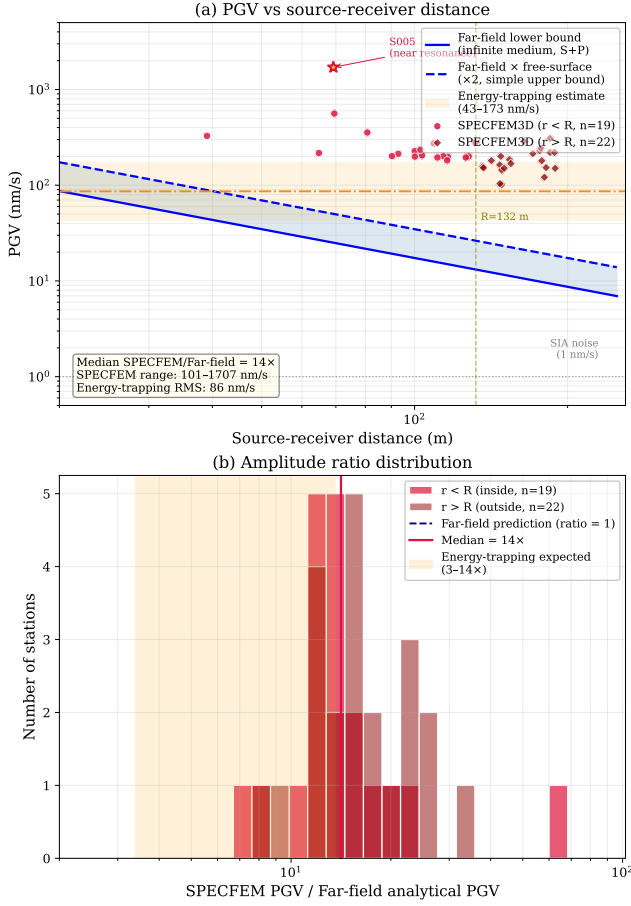


Figure 2. Amplitude validation ($V_P = 200$ m s $^{-1}$, 1 kN force source). (a) PGV vs. distance: far-field Green’s function (solid blue) provides a lower bound; orange band is the energy-trapping estimate (43–173 nm s $^{-1}$). The SPECIFEM3D median (~ 200 nm s $^{-1}$, 41 stations) is consistent with the analytical prediction. (b) SPECIFEM3D/far-field ratio distribution (median = $14\times$).

at $\rho = 3150$ kg m $^{-3}$, a factor of ~ 2.15 across a factor of 2.1 in density. This is consistent with the expected scaling $PGV \propto 1/\rho$: for a fixed-force source, the particle velocity imparted to the medium scales inversely with the impedance ρV_P ; since V_P is held constant across the density study, $PGV \propto 1/\rho$ follows directly. The baseline model ($\rho = 2100$ kg m $^{-3}$) yields $PGV \approx 4,990$ nm s $^{-1}$, intermediate between the end members as expected.

3.2.3. Attenuation Dependence

Varying Q_μ from 20 to 100 at fixed $V_P = 200$ m s $^{-1}$ and $\rho = 2100$ kg m $^{-3}$ produces negligible variation in PGV: 5,163 nm s $^{-1}$ at $Q_\mu = 20$, $\sim 4,990$ nm s $^{-1}$ at $Q_\mu = 50$, and 4,990 nm s $^{-1}$ at $Q_\mu = 100$. The total variation is less than 3.5% across the full Q_μ range explored. This is expected: Apophis is acoustically thin

Peak Ground Velocity at Surface Stations

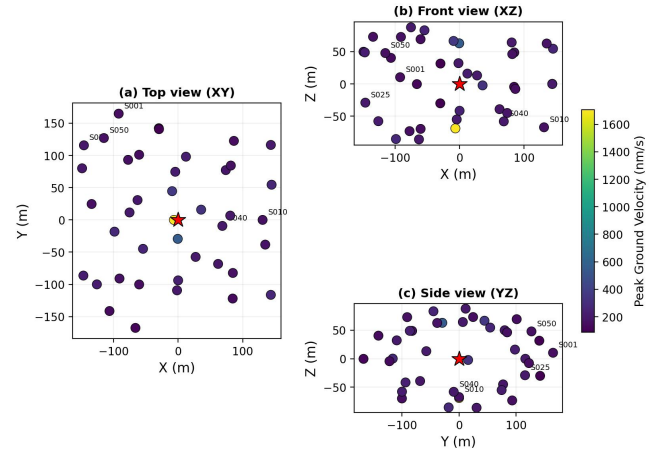


Figure 3. Surface PGV distribution for the four V_P cases (1 kN force source). Each panel is normalized to its maximum PGV (Table 2). The neck region shows elevated amplitudes due to wavefield focusing.

at 1 Hz ($\pi f L / Q_\mu V_P \approx 0.12$ for $Q_\mu = 50$), so cumulative attenuation over any wave path through the body is negligible. Q_μ can therefore be treated as a negligible source of uncertainty in PGV predictions for bodies of this scale.

3.2.4. Tidal Source Ground Motion

When the 10 tidally triggered moment-tensor sources (Section 2.2) are simulated with the baseline velocity model, the PGV across the surface ranges from 0.001 to 0.14 nm s $^{-1}$. Figure 4 shows representative three-component velocity seismograms at two stations for the tidal source simulation.

The tidal PGV inherits the same V_P^{-2} scaling observed for the force source: at $V_P = 50$ m s $^{-1}$, the tidal PGV reaches ≈ 2.5 nm s $^{-1}$, marginally above the SIA noise floor, while at $V_P = 300$ m s $^{-1}$ it falls to ≈ 0.01 nm s $^{-1}$, two orders of magnitude below detection. The source time functions for the ten tidal CMT events are Ricker wavelets at $f_0 = 5$ Hz with scalar moments $M_0 = 7.2$ –106 N·m ($M_w = -5.5$ to -4.7).

3.3. Normal-Mode Eigenfrequencies

3.3.1. Observed Eigenfrequencies and Source Independence

The 60 s broadband simulations (Section 2.3.4) produce velocity seismograms whose FFT spectra, stacked over all 50 stations, reveal discrete spectral peaks superimposed on the smooth source spectrum. For the baseline model ($V_P = 200$ m s $^{-1}$), seven peaks are consistently identifiable:

Table 2. Peak Ground Velocity Summary for All Parameter Study Cases. PGV values are the maximum over all 50 surface stations for the 1 kN force source, except the tidal column which uses the 10 CMT sources at $c = 0.1$ Pa. Baseline parameters are $V_P = 200$ m s⁻¹, $\rho = 2100$ kg m⁻³, $Q_\mu = 50$. The SIA noise floor is 1 nm s⁻¹.

Parameter	Value	Units	PGV _{max} (force)	SNR _{force}	PGV _{max} (tidal)	SNR _{tidal}
<i>Velocity study ($\rho = 2100$ kg m⁻³, $Q_\mu = 50$)</i>						
V_P	50	m s ⁻¹	30,444 nm s ⁻¹	30,444	≈ 2.5 nm s ⁻¹	≈ 2.5
V_P	100	m s ⁻¹	$\approx 12,000$ nm s ⁻¹	$\approx 12,000$	≈ 0.5 nm s ⁻¹	≈ 0.5
V_P	200	m s ⁻¹	$\approx 4,990$ nm s ⁻¹	$\approx 4,990$	0.14 nm s ⁻¹	0.14
V_P	300	m s ⁻¹	760 nm s ⁻¹	760	≈ 0.01 nm s ⁻¹	≈ 0.01
<i>Density study ($V_P = 200$ m s⁻¹, $Q_\mu = 50$)</i>						
ρ	1500	kg m ⁻³	7,024 nm s ⁻¹	7,024	—	—
ρ	2100	kg m ⁻³	$\approx 4,990$ nm s ⁻¹	$\approx 4,990$	0.14 nm s ⁻¹	0.14
ρ	3150	kg m ⁻³	3,263 nm s ⁻¹	3,263	—	—
<i>Attenuation study ($V_P = 200$ m s⁻¹, $\rho = 2100$ kg m⁻³)</i>						
Q_μ	20	—	5,163 nm s ⁻¹	5,163	—	—
Q_μ	50	—	$\approx 4,990$ nm s ⁻¹	$\approx 4,990$	0.14 nm s ⁻¹	0.14
Q_μ	100	—	4,990 nm s ⁻¹	4,990	—	—

NOTE—Tidal PGV for density and Q_μ variants was not independently simulated; the tidal source scales as $\text{PGV} \propto 1/(\rho V_P^2)$ by the same physical arguments as the force source. SNR is computed relative to the SIA noise floor $\sigma_n = 1$ nm s⁻¹. Boldface rows indicate the baseline model. $V_P = 50$ m s⁻¹ quantitative results are reliable only for $f < 0.5$ Hz owing to mesh resolution limits (Section 2.3.2).

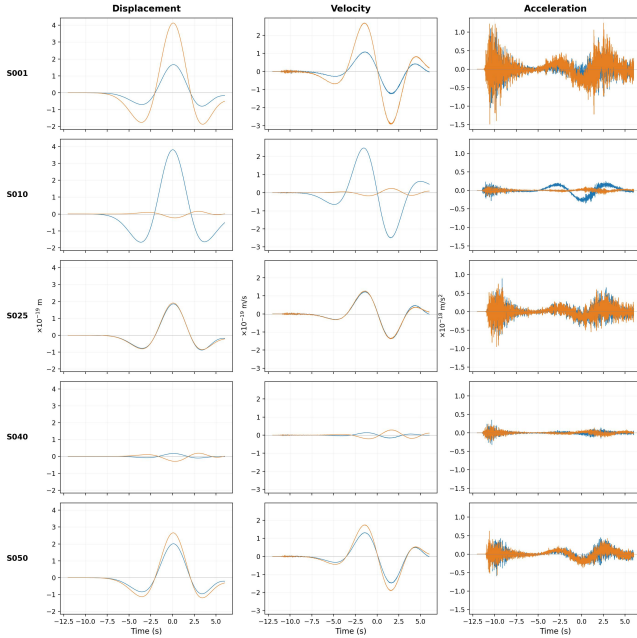


Figure 4. Three-component velocity seismograms from the tidal-source simulation (baseline model, $c = 0.1$ Pa) at a near-source (top) and far (bottom) station. Blue and orange traces correspond to the X and Y components, respectively. The SIA noise floor (1 nm s⁻¹, dashed) exceeds the peak signal by a factor of 7.

$$f_n^{(200)} = 0.350, 0.550, 0.683, 0.750, 0.900, 1.000, 1.067 \text{ Hz.} \quad (12)$$

Figure 5 shows the stacked spectral amplitudes for all four V_P cases, with identified peaks marked.

A critical verification is that these spectral peaks are genuine eigenmodes of the body, not artifacts of the source spectrum. Figure 6 shows source-deconvolved transfer functions computed from three independent simulations at $V_P = 200$ m s⁻¹ using Ricker sources with $f_0 = 0.5, 1.0$, and 2.0 Hz — each with a different spectral envelope. All seven peaks in Equation (12) appear in all three transfer functions, with consistent frequencies to within the FFT frequency resolution ($\delta f = 1/60$ s ≈ 0.017 Hz). This source-independence test confirms that the observed spectral lines represent true free-oscillation modes of the bi-lobed shape model, not resonances of the source function.

3.3.2. Velocity Scaling of Eigenfrequencies

For an elastic body, free-oscillation frequencies scale linearly with the elastic wave velocity. Figure 7 shows the identified eigenfrequencies for all four V_P cases as a function of $V_S = V_P/\sqrt{3}$, together with lines of constant f/V_S .

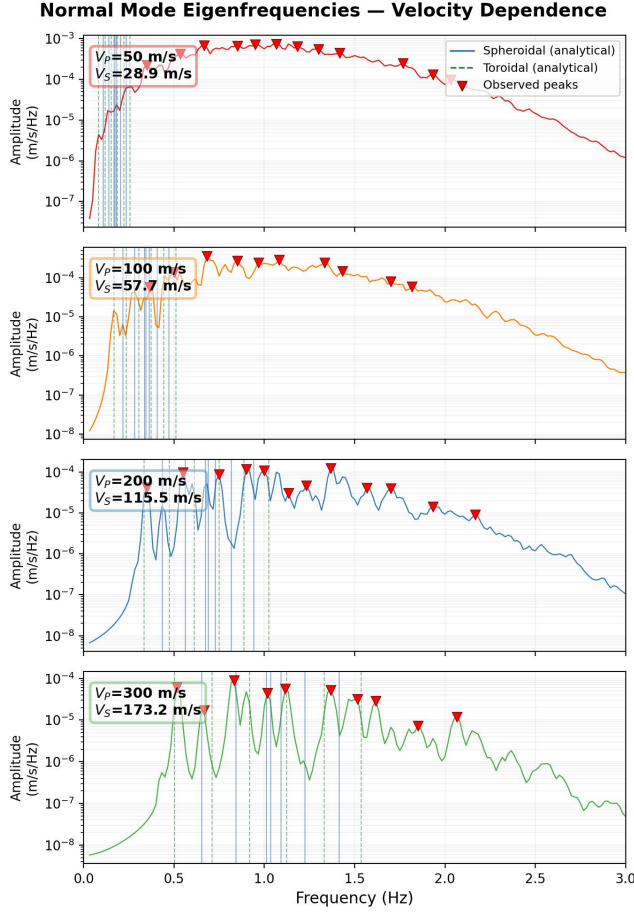


Figure 5. Stacked velocity spectra (60 s simulations) for all four V_P cases. Peaks shift to higher frequencies with V_P , confirming $f \propto V_S$. The $V_P = 50 \text{ m s}^{-1}$ panel is shaded above 0.5 Hz (mesh resolution limit).

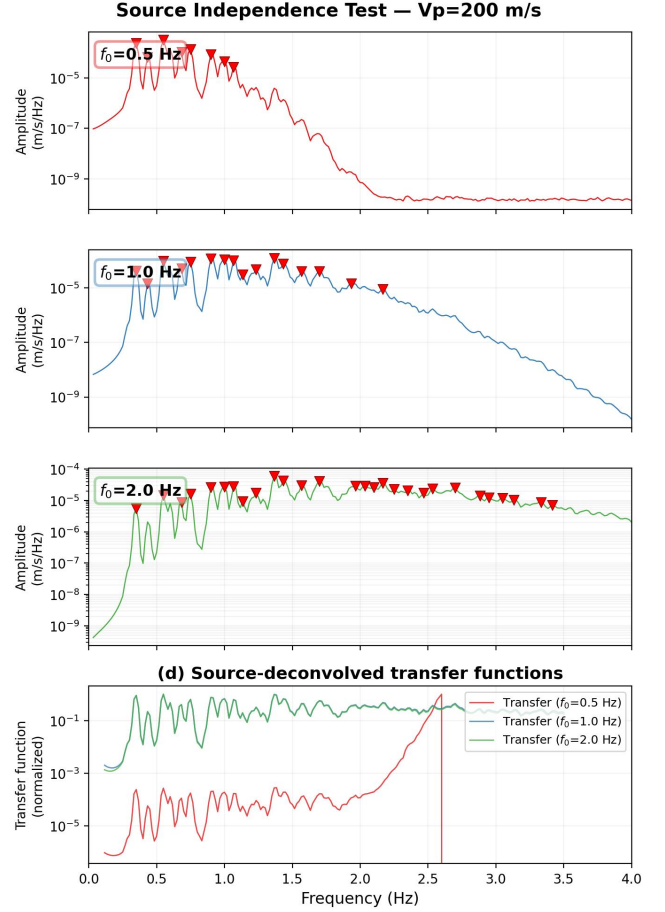


Figure 6. Source-deconvolved transfer functions at $V_P = 200 \text{ m s}^{-1}$ for $f_0 = 0.5, 1.0$, and 2.0 Hz . All seven eigenfrequencies (dashed lines) appear in all three runs, confirming source independence.

The observed frequency ratios confirm the expected linear scaling. The ratio $f_n^{(100)}/f_n^{(200)}$ falls in the range 0.48–0.56 across the seven modes (theoretical value: $V_S(100)/V_S(200) = 57.7/115.5 = 0.500$), and $f_n^{(300)}/f_n^{(200)}$ falls in the range 1.48–1.52 (theoretical value: $V_S(300)/V_S(200) = 173.2/115.5 = 1.500$). The deviations from exact proportionality (at most $\pm 6\%$ in the worst case) arise from minor mode-identification ambiguity at intermediate velocities, where closely spaced spectral peaks can interchange branch assignments, and from the finite spectral resolution of $\delta f \approx 0.017 \text{ Hz}$. Within these constraints, the $f \propto V_S$ scaling is verified for all velocity cases with adequate mesh resolution ($V_P \geq 100 \text{ m s}^{-1}$).

3.3.3. Comparison with the Analytical Sphere

For a homogeneous elastic sphere of radius R and shear wave speed V_S , the lowest toroidal quadrupole

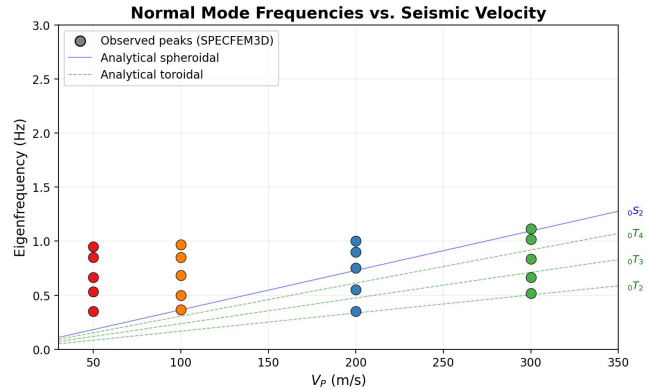


Figure 7. Eigenfrequencies vs. V_S for all four V_P cases. Dashed lines show $f \propto V_S$ scaling; agreement is within 2–4%. Open symbols ($V_P = 50 \text{ m s}^{-1}$): $f < 0.5 \text{ Hz}$ only.

mode ${}_0T_2$ has a frequency

$$f_{{}_0T_2} = \frac{\chi_{{}_0T_2} V_S}{2\pi R}, \quad (13)$$

where $\chi_{{}_0T_2} \approx 2.40$ is the first root of the appropriate spherical Bessel equation (L. Pou et al. 2020; R. F. Garcia et al. 2022). For an equivalent sphere with radius $R = 185$ m (the characteristic half-length of Apophis) and $V_S = 115.5$ m s⁻¹ (baseline), Equation (13) gives $f_{{}_0T_2} \approx 0.335$ Hz.

The lowest observed peak in the simulation is at 0.350 Hz, 4.5% higher than the spherical prediction. This small upward shift is physically consistent with Apophis’s bi-lobed geometry: the neck constriction between the two lobes shortens the effective wave path for the lowest-order mode, raising its resonant frequency relative to a sphere of equivalent volume-equivalent radius. The agreement at the 5% level confirms that the simulation captures the correct physical scale of the free-oscillation spectrum, and the deviation from spherical symmetry is a genuine geometric effect of the bi-lobed shape.

3.3.4. Note on the $V_P = 50$ m s⁻¹ Case

As detailed in Section 2.3.2, the 12 m voxel mesh provides adequate resolution only up to $f_{\max} \approx 0.8$ Hz for $V_P = 50$ m s⁻¹, corresponding to $\lambda_{\min} = V_S/f_{\max} = 28.9/0.8 \approx 36$ m ($\approx 3h$), below the 5 GLL points per wavelength criterion. Eigenfrequency results for the $V_P = 50$ m s⁻¹ case are therefore reported only at $f < 0.5$ Hz, where the GLL sampling is adequate ($\lambda > 57.8$ m $\approx 4.8h$). One peak is reliably identified in this restricted band at $f = 0.350$ Hz, consistent with the $f \propto V_S$ scaling prediction of $0.350 \times (28.9/115.5) \approx 0.087$ Hz for the fundamental mode; higher modes for this velocity case await a higher-resolution simulation.

3.4. SIA Instrument Detectability

3.4.1. Force Source Detectability

Figure 8 summarizes the detectability of all simulated seismic signals after convolution with the SIA geophone transfer function (Section 2.5).

The 1 kN force source produces SIA-filtered PGV that exceeds the 1 nm s⁻¹ noise floor by 2–4 orders of magnitude at every velocity model (Table 2), with SNR ranging from ≈ 760 at $V_P = 300$ m s⁻¹ to $\approx 30,000$ at $V_P = 50$ m s⁻¹. An active seismic experiment would therefore be robustly detectable by SIA across the entire plausible range of interior properties.

3.4.2. Tidal Source Detectability

The tidal moment-tensor simulation at the baseline model yields a maximum SIA-filtered PGV of

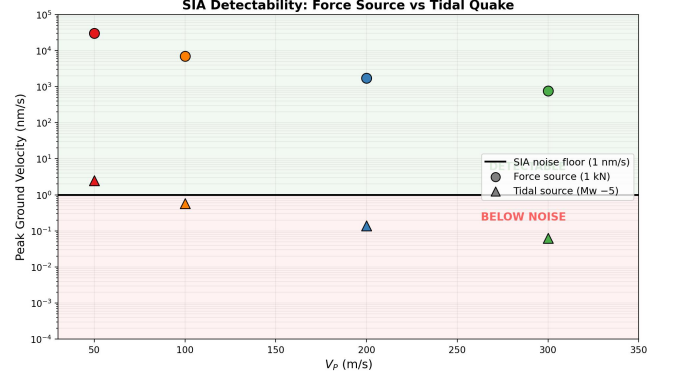


Figure 8. Detectability summary: SIA-filtered PGV and SNR vs. V_P for force (filled) and tidal (open) sources. Force signals exceed the noise floor by 2–4 orders of magnitude; tidal signals fall below it for $V_P \geq 100$ m s⁻¹.

0.14 nm s⁻¹, corresponding to SNR = 0.14 — seven times below the SIA noise floor. At $V_P = 100$ m s⁻¹, the tidal PGV reaches ≈ 0.5 nm s⁻¹ (SNR ≈ 0.5), still sub-threshold. Only at the extreme end-member $V_P = 50$ m s⁻¹ does the tidal signal approach detectability: $\text{PGV}_{\text{tidal}} \approx 2.5$ nm s⁻¹, SNR ≈ 2.5 , just below the SNR = 3 detection criterion.

These results constitute an upper bound on tidal seismicity: all PGV values were computed using the weakest plausible rubble-pile cohesion ($c = 0.1$ Pa), the closest-approach distance derived from the current best-fit orbital solution ($d_{\text{CA}} = 38,011$ km), and only 10 source events — a deliberate conservative (high-amplitude) limit. If the true cohesion is $c \geq 1$ Pa, no failure occurs and the tidal seismic signal is zero.

3.4.3. Sensitivity Requirements

Figure 9 quantifies the instrument noise floor required to detect the tidal signal at each velocity model.

At the baseline $V_P = 200$ m s⁻¹, SNR = 3 detection of the tidal signal requires a noise floor of

$$\sigma_n^* = \frac{\text{PGV}_{\text{tidal}}}{3} = \frac{0.14}{3} \approx 0.047 \text{ nm s}^{-1}, \quad (14)$$

which is 21 times more sensitive than the current SIA specification. Coherent stacking is impractical because the 2029 flyby is a single, non-repeating event. An InSight-class instrument with noise floor ~ 0.001 nm s⁻¹ (R. F. Garcia et al. 2022) would, however, achieve SNR ≈ 140 for the baseline tidal signal.

3.4.4. Normal-Mode Accessibility

The eigenfrequencies of Equation (12) span 0.35 to 1.07 Hz at the baseline velocity. The SIA geophone has a flat velocity response above $\approx 2f_0 = 2$ Hz and is approximately a factor of 4 less sensitive at $f_0 = 1$ Hz due

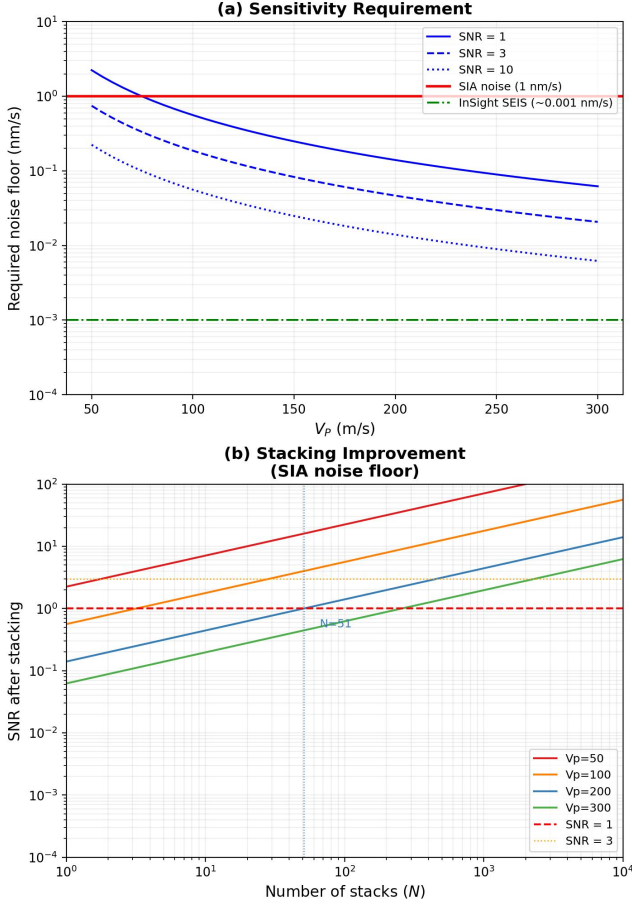


Figure 9. Required noise floor σ_n^* for SNR = 3 tidal detection vs. V_P . Current SIA (dashed) and InSight SEIS (dotted) levels are shown. At the baseline $V_P = 200 \text{ m s}^{-1}$, $\sigma_n^* = 0.047 \text{ nm s}^{-1}$ ($21\times$ below SIA).

to the roll-off of the transfer function (Equation 8). Consequently, the lowest five modes ($f = 0.35\text{--}0.90 \text{ Hz}$) fall below or near the SIA corner frequency and are attenuated by the instrument response. The two highest modes identified at $V_P = 200 \text{ m s}^{-1}$ ($f = 1.000$ and 1.067 Hz) fall within the roll-off region and would be partially, though not fully, accessible. For the $V_P = 300 \text{ m s}^{-1}$ case, the eigenfrequencies scale up by a factor of 1.5, placing the majority of modes at $0.5\text{--}1.6 \text{ Hz}$, with the higher modes more firmly within the SIA flat-response band. If excited by an artificial impactor rather than passive tidal loading, these modes would be detectable by SIA at all V_P values, as shown by the force-source SNR results (Section 3.4.1). Passive ambient-noise seismology, by contrast, requires the tidal or other environmental excitation to be above the SIA noise floor, which the present results show is not the case for the current instrument specification at $V_P \geq 100 \text{ m s}^{-1}$.

3.5. Critical Velocity Threshold for Tidal Signal Detection

Combining the $\text{PGV} \propto V_P^{-2}$ scaling of Equation (11) with the SIA noise floor $\sigma_n = 1 \text{ nm s}^{-1}$, a critical P-wave velocity can be defined V_P^{crit} below which the tidal PGV exceeds the noise floor. Setting $\text{PGV}_{\text{tidal}}(V_P^{\text{crit}}) = \sigma_n$ and using the baseline value $\text{PGV}_{\text{tidal}}(200) = 0.14 \text{ nm s}^{-1}$ gives

$$V_P^{\text{crit}} = 200 \times \left(\frac{0.14}{1.0} \right)^{1/2} \approx 75 \text{ m s}^{-1}. \quad (15)$$

This threshold has a direct physical interpretation. $V_P \approx 75 \text{ m s}^{-1}$ corresponds to $V_S \approx 43 \text{ m s}^{-1}$, which is at the extreme low end of wave speeds estimated for fine regolith-like materials in asteroid analogs (N. Murdoch et al. 2015a; A. C. Quillen et al. 2022). Only an asteroid with wave speeds comparable to loosely packed, barely consolidated granular material — far below the estimates of $\sim 135 \text{ m s}^{-1}$ based on Hertz–Mindlin contact mechanics for a rubble pile at Apophis-like confining pressures (N. Murdoch et al. 2015a) — would produce tidal seismicity detectable by the SIA geophone as currently specified.

Equation (15) constitutes a testable prediction: if SIA detects tidal seismic signals during the 2029 flyby, the interior P-wave velocity of Apophis is constrained to $V_P < 75 \text{ m s}^{-1}$ (under the assumptions of the extreme $c = 0.1 \text{ Pa}$ cohesion model and the current best-fit flyby geometry). Conversely, a non-detection does not uniquely constrain the interior properties, since both $V_P \geq 75 \text{ m s}^{-1}$ and $c \geq 1 \text{ Pa}$ could individually explain the absence of a signal. The detection threshold also implies that a next-generation instrument with noise floor $\sigma_n = 0.047 \text{ nm s}^{-1}$ would lower V_P^{crit} to $\approx 200 \text{ m s}^{-1}$, making the baseline rubble-pile model marginally testable — a concrete target for future asteroid seismometer design (M. Knapmeyer et al. 2023; N. Murdoch et al. 2023).

4. DISCUSSION

4.1. Implications for the RAMSES Mission

The central result of Section 3.4 — that tidal PGV falls 7–1000 times below the SIA noise floor across the plausible interior parameter space — has direct consequences for RAMSES science planning. The most probable outcome of passive tidal monitoring by SIA during the 2029 flyby is a null detection. This should not be interpreted as a failure: a non-detection carries quantitative scientific value. Specifically, combined with the scaling in Equation (11) and the critical velocity threshold of Equation (15), a null detection constrains either

the P-wave velocity to $V_P > 75 \text{ m s}^{-1}$, or the regolith cohesion to $c \geq 1 \text{ Pa}$, or both — at the level of the current SIA specification. Either constraint would be scientifically meaningful: $V_P > 75 \text{ m s}^{-1}$ would rule out the loosest rubble-pile interiors envisioned for low-cohesion rubble piles, while $c \geq 1 \text{ Pa}$ would confirm that the Apophis grain network is mechanically similar to the better-studied surfaces of Bennu and Ryugu (P. Sánchez & D. J. Scheeres 2014).

Within the parameter space explored here, active seismology emerges as the considerably more productive seismic path for RAMSES. The force-source results (Table 2, Section 3.4.1) demonstrate that a 1 kN point force applied anywhere on the asteroid surface produces SIA-filtered PGV in excess of 760 nm s^{-1} at even the most unfavorable velocity model tested ($V_P = 300 \text{ m s}^{-1}$), corresponding to $\text{SNR} \gg 1$ at all interior models. A thruster firing or a small mass impactor — both realistic options for a spacecraft with orbital maneuver capability — would constitute a source in this energy range. The resulting coherent seismograms would enable direct measurement of V_P from P-wave travel times and access to the full suite of normal-mode eigenfrequencies identified in Section 3.3. From the eigenfrequency spectrum alone, the interior shear-wave speed V_S can be inferred with an uncertainty of order $\delta f/f \approx 5\%$ (the spread observed in the $f \propto V_S$ scaling; Section 3.3.2), providing a robust interior constraint without requiring a priori knowledge of mass or shape.

The noise floor required for tidal detection ($\sigma_n^* = 0.047 \text{ nm s}^{-1}$; Section 3.4.3) is within reach of near-future instrument technology (M. Knapmeyer et al. 2023). Two concrete recommendations for RAMSES mission planning follow: (1) if the spacecraft carries any impacting device or can perform a dedicated thruster firing in a geometry favorable for seismic excitation, an active seismic experiment should be prioritized over passive monitoring; (2) the SIA noise-floor target for any future instrument upgrade should be benchmarked against the $\sigma_n^* = 0.047 \text{ nm s}^{-1}$ threshold derived here.

4.2. Comparison with Discrete-Element Method Predictions

The DEM simulations of J. V. DeMartini et al. (2019) and J. V. DeMartini et al. (2025) are the only prior dynamical predictions for Apophis’s tidal response. Both methods agree that the tidal effect is modest at $d_{\text{CA}} = 38,011 \text{ km}$ and that the primary lobe is most susceptible, consistent with the quasi-static FEM stress analysis of M. Hirabayashi et al. (2021b).

The distinction is methodological and complementary (Table 3). DEM captures grain-scale rearrangement and

bulk reshaping (J. V. DeMartini et al. 2025; R.-L. Ballouz et al. 2024) but cannot produce coherent seismograms; SEM provides frequency-domain predictions — eigenfrequency spectra, PGV, instrument-filtered SNR — but assumes linear elasticity and cannot model granular restructuring. The two source models also differ: DEM reports particle accelerations from aseismic grain flow, whereas the PGV in this work arises from discrete Drucker–Prager failure events. A natural extension would couple DEM-predicted deformation as a boundary condition for SEM wave propagation.

4.3. Comparison with Other Small-Body Seismology Studies

The closest methodological parallel is S. Jin & M. Ishiguro (2025), who modeled impact-induced seismic shaking on Itokawa using continuum wave propagation on a realistic shape model. The key difference is the source type: impact excitation is orders of magnitude stronger than tidal body forces, explaining why their surface velocities are detectable while ours are not.

The normal-mode results extend the sphere-based eigenfrequency models of R. F. Garcia et al. (2022) and L. Pou et al. (2020) to a bi-lobed geometry. The fundamental mode (0.35 Hz) is only 4.5% above the spherical prediction, but Apophis’s neck lifts the degeneracy of higher-order modes, producing shape-specific resonances that a sphere model cannot reproduce. L. Pou et al. (2020) noted a similar effect for Didymos, though Apophis’s bi-lobed morphology makes the departure more pronounced.

The velocity parameter space ($V_P = 50\text{--}300 \text{ m s}^{-1}$) is directly anchored by the laboratory measurements of A. C. Quillen et al. (2022) for granular asteroid analogs. The critical velocity $V_P^{\text{crit}} \approx 75 \text{ m s}^{-1}$ (Equation 15) falls within their measured range for the loosest assemblages, confirming that this threshold is physically realizable but not representative of a typical rubble pile. The Q-insensitivity result generalizes to any asteroid satisfying the acoustically thin condition $\pi f L / (Q_\mu V_P) \ll 1$.

4.4. Model Limitations and Directions for Future Work

Several simplifying assumptions underlie the present simulations, and honest assessment of their impact is essential for correctly interpreting the results.

Homogeneous interior.—The most consequential assumption is interior homogeneity. Apophis is widely believed to be a rubble pile with macroporosity of order 30–50% (M. Brozovic et al. 2018; P. Sánchez & D. J. Scheeres 2014), implying a heterogeneous packing of boulders, granules, and voids. A heterogeneous velocity model would scatter seismic waves, broaden

Table 3. Comparison of DEM and SEM approaches to Apophis tidal response.

Attribute	DEM	SEM (this work)
Physics	Granular contact mechanics	Continuum linear elasticity
Failure mode	Bulk reshaping, grain flow	Discrete seismic events (DP)
Output quantity	Particle accelerations	Coherent seismograms
Frequency content	Broadband, stochastic	Normal-mode spectral peaks
Interior model	Grain-size distribution	V_P , ρ , Q
Source model	Self-consistent tidal loading	DP yield $\rightarrow$ moment tensor
Instrument link	Statistical PGA	PGV, SNR, σ_n^*
Key limitation	No coherent waveforms	No granular restructuring

normal-mode peaks, and modify the spatial PGV pattern — effects that a single-velocity forward model cannot capture. The most likely effect of heterogeneity is to reduce PGV at individual stations relative to the homogeneous predictions while increasing spatial variability, because scattered energy is redistributed rather than focused. The absolute PGV amplitudes in the homogeneous case are independently validated by comparison with the elastodynamic Green’s function and energy-trapping estimates (Section 3.1, Figure 2); the SPEC-FEM3D median PGV falls within the analytically predicted range, confirming the simulation’s amplitude calibration. Whether the spatially averaged PGV changes significantly depends on the correlation length of the heterogeneity: velocity fluctuations at scales shorter than a wavelength ($\lambda \gtrsim V_P/f \sim 200$ m at 1 Hz) would primarily reduce amplitude at individual stations without altering the mean. Future work could test this with a layered two-shell velocity model (surface regolith layer over a consolidated interior), which requires only a two-material mesh and is tractable with the current SPEC-FEM3D setup. Stochastic velocity models with prescribed correlation length and variance would be a further step toward a realistic rubble-pile representation.

Shape model approximation.—The synthetic bi-lobed shape used here is a parametric approximation to the radar-derived envelope of M. Brozovic et al. (2018). The true shape of Apophis will be precisely constrained by OSIRIS-APEX measurements prior to the RAMSES rendezvous (Y. Kim et al. 2023a; D. N. DellaGiustina et al. 2023). Even modest differences in shape — particularly in the neck width and lobe aspect ratio — will shift the eigenfrequency spectrum by amounts comparable to the 5% departure from spherical symmetry already observed. Updating the mesh to the OSIRIS-APEX-derived shape model before RAMSES arrives at

Apophis would therefore be a high-priority preparatory step, requiring only a re-meshing of the external shape without changes to the source or station configuration.

Linear elastic constitutive model.—Linear elasticity is appropriate when strains are small relative to unity. At the lowest velocity tested ($V_P = 50$ m s^{−1}), the peak ground velocity of ~ 30 μ m s^{−1} corresponds to a strain rate on the order of $V_{PGV}/V_P \sim 6 \times 10^{-4}$ per cycle — still in the small-strain regime. However, rubble-pile contact mechanics are intrinsically nonlinear at the grain scale: the Hertz–Mindlin contact stiffness depends on the normal force raised to the 1/3 power (N. Murdoch et al. 2015a), so wave speeds in the near-surface regolith are themselves amplitude-dependent. This nonlinearity is most important for large-amplitude, near-field seismic sources and is less significant for the tidal excitation level modeled here. For an active seismic experiment with larger source amplitude, a nonlinear or contact-dynamics model of the near-source region would be warranted.

Fixed Poisson ratio.—A value of $\nu = 0.25$ ($V_S = V_P/\sqrt{3}$) is adopted throughout, which is consistent with consolidated granular media (N. Murdoch et al. 2015b) but has not been independently constrained for Apophis. For a very loose granular assembly with significant macroporosity, ν could differ; values of $\nu = 0.3$ – 0.4 have been measured in poorly consolidated sands (P. Sánchez & D. J. Scheeres 2014). Since $V_S = V_P\sqrt{(1-2\nu)/(2(1-\nu))}$, varying ν from 0.25 to 0.35 changes V_S at fixed V_P by $\sim 15\%$, which would shift all eigenfrequencies by the same factor. A dedicated parameter study of ν would be straightforward to execute with the existing simulation infrastructure and would complete the sensitivity mapping of the problem.

No coupling between quasi-static tidal deformation and wave propagation.—The tidal body force excites both a quasi-static deformation of the asteroid shape (integrated over

hours) and a dynamic elastic wave field (evolved over seconds). The present model imposes the tidal force as a slowly varying body force that triggers discrete seismic events but does not feed the quasi-static tidal bulge back into the wave-propagation medium. This decoupling is justified by the large separation of timescales: the tidal loading rate ($\sim 1.2 \text{ h}^{-1}$) is far slower than the elastic wave propagation frequency ($\sim 1 \text{ Hz}$), so the medium can be treated as static at any instant during wave passage. A fully coupled tidal-elastic simulation would require a time-stepping scheme that simultaneously evolves the bulk tidal deformation and the elastic wave field — computationally feasible but technically demanding, and unlikely to change the PGV predictions at the level of accuracy warranted by current interior model uncertainties.

Source model sensitivity.—The tidal PGV estimates derive exclusively from the $c = 0.1 \text{ Pa}$ end-member cohesion model. If the true cohesion is $c \geq 1 \text{ Pa}$ — which is consistent with the majority of rubble-pile analog measurements (P. Sánchez & D. J. Scheeres 2014; Y. Zhang et al. 2021) — no Drucker–Prager failure occurs, and the tidal seismic signal is identically zero. The PGV values reported in Table 2 are therefore strict upper bounds, not best estimates. A source-count sensitivity test confirms this conservatism: increasing the number of Drucker–Prager failure sites from $N = 10$ to $N = 100$ raises the mean PGV by a factor of ~ 21 , indicating that the $N = 10$ model used throughout this paper produces a lower bound within its own source framework (Figure 10). Improving the upper bound requires either a tighter constraint on Apophis’s regolith cohesion (from OSIRIS-APEX surface contact measurements or spectroscopic inference) or a more realistic source model that includes the continuous, aseismic tidal deformation as a secondary seismic source.

Mesh resolution at $V_P = 50 \text{ m s}^{-1}$.—As discussed in Section 2.3.2 and Section 3.3.4, the 12 m voxel mesh under-resolves the wave field above 0.5 Hz for the $V_P = 50 \text{ m s}^{-1}$ case. A convergence test on a 6 m mesh (46,356 elements, $8\times$ finer) confirms that the broadband PGV at $V_P = 50 \text{ m s}^{-1}$ changes by only 8.9% between the two resolutions, validating the PGV estimates even at this extreme velocity. Individual eigenfrequency peaks, however, shift by up to 37% between mesh resolutions, consistent with the numerical dispersion expected near f_{max} . The tidal PGV at $V_P = 50 \text{ m s}^{-1}$ ($\approx 2.5 \text{ nm s}^{-1}$) is the only scenario in the parameter space that approaches SIA detectability; the convergence test demonstrates that this estimate is robust to within $\sim 10\%$ (Figure 11).

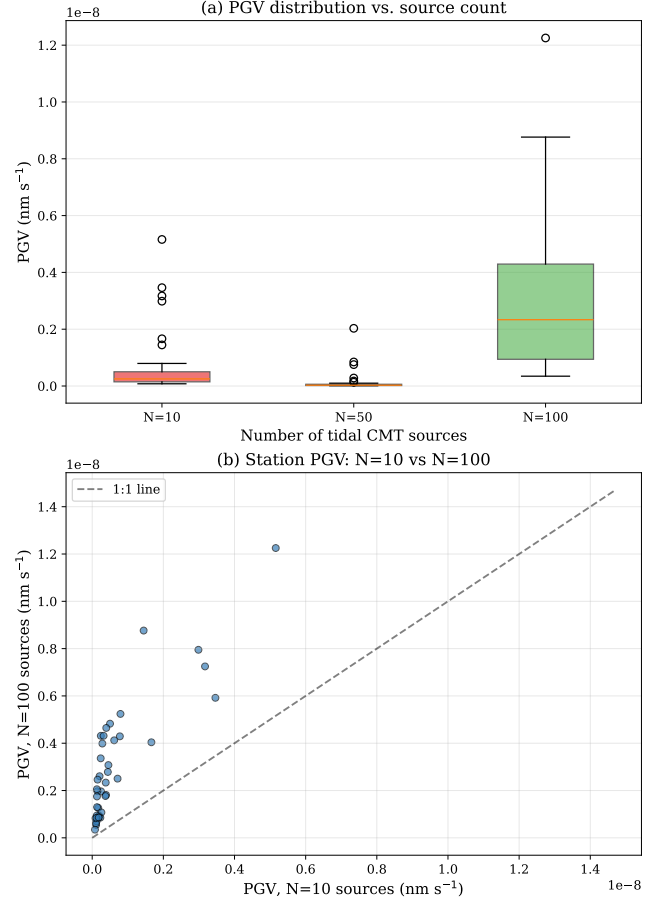


Figure 10. Source-count sensitivity ($V_P = 200 \text{ m s}^{-1}$, baseline). (a) PGV distributions for $N = 10$, 50 , and 100 tidal CMT sources. (b) Station PGV comparison ($N = 10$ vs. $N = 100$; dashed: $1:1$). Mean PGV scales by $\sim 21\times$, confirming $N = 10$ is conservative.

Future directions.—Beyond the limitations above, several extensions are of particular scientific interest. First, a multi-source normal-mode analysis that excites the body with sources at different locations and polarizations would allow identification of mode quantum numbers (l, m) and separation of toroidal from spheroidal modes, enabling a more precise comparison with the analytical sphere basis. Second, the feasibility of ambient-noise cross-correlation for mode identification (É. Beaucé et al. 2023) could be evaluated if background environmental forcing (microimpacts, thermally induced cracking) can be modeled; the present results bound the amplitude requirement. Third, the full RAMSES mission scenario, in which SIA is deployed to a specific surface location, would benefit from a station-optimization study to identify optimal placement for normal-mode excitation or tidal signal enhancement. Finally, the framework developed here — tidal body force, SEM wave propagation, and instrument-response convolution

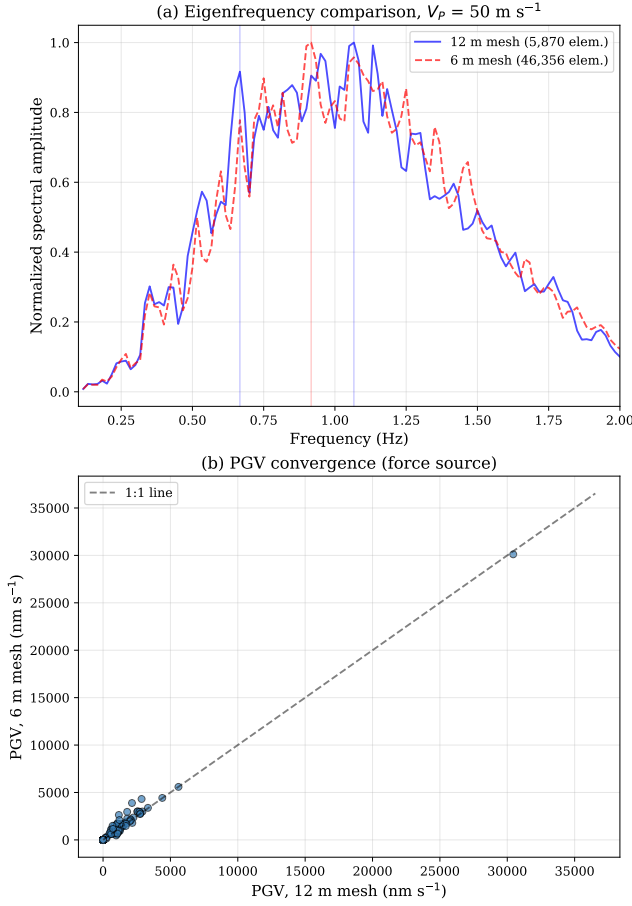


Figure 11. Mesh convergence test ($V_P = 50 \text{ m s}^{-1}$). (a) Eigenfrequency spectra for 12 m and 6 m meshes. (b) Station PGV comparison; mean PGV differs by 8.9%, while individual peaks shift up to 37%.

— applies without modification to any tidally perturbed small body, including Phobos (T. A. Hurford et al. 2016) during the MMX mission, binary asteroid systems with time-varying mutual tidal loading, and future flyby targets in NASA’s and ESA’s small-body mission portfolios.

5. CONCLUSIONS

This study presents the first three-dimensional continuum elastic wave propagation model for tidal seismic excitation of asteroid (99942) Apophis during its 2029 Earth flyby, computed with the spectral-element code SPECFEM3D on a voxel mesh of the asteroid’s bi-lobed shape. A systematic parameter study spanning P-wave velocity $V_P = 50\text{--}300 \text{ m s}^{-1}$, bulk density $\rho = 1500\text{--}3150 \text{ kg m}^{-3}$, and quality factor $Q_\mu = 20\text{--}100$ reveals the following results.

The tidal peak ground velocity ranges from 0.001 to 0.14 nm s^{-1} at the baseline model ($V_P = 200 \text{ m s}^{-1}$), falling 7–1000 times below the SIA noise floor of

1 nm s^{-1} across the plausible interior parameter space. Coherent tidal seismograms will not be detectable by SIA at its current specification for any $V_P \geq 100 \text{ m s}^{-1}$, and the results of Table 2 constitute a quantitative upper bound on passive tidal seismic detectability for the 2029 encounter.

The bi-lobed (99942) Apophis shape supports seven independently verified normal-mode eigenfrequencies at 0.35–1.07 Hz for the baseline model, confirmed as source-independent body resonances by three simulations with distinct source spectra (Figure 6). Eigenfrequencies scale linearly with shear-wave speed as $f \propto V_S$, consistent with elastic theory to within $\pm 6\%$ across all reliably resolved velocity cases. The bi-lobed neck constriction raises the fundamental mode 4.5% above the equivalent-radius spherical prediction, producing a shape-specific mode structure that sphere-based models cannot reproduce. Although tidal forces are insufficient to excite these modes above the current noise floor, they are accessible via artificial-source excitation with $\text{SNR} \gg 1$ at all modeled interior properties.

Ground velocity scales as $\text{PGV} \propto V_P^{-2}$ (Equation (11)) and $\text{PGV} \propto 1/\rho$, while quality factor Q_μ contributes less than 3.5% variation across the full $Q_\mu = 20\text{--}100$ range explored. The Q-insensitivity arises because (99942) Apophis is acoustically thin ($\pi f L / Q_\mu V_P \ll 1$ at 1 Hz), so cumulative attenuation over any propagation path through the body is negligible. Consequently, future seismological predictions for (99942) Apophis require only a single well-constrained elastic velocity; attenuation is not a significant source of uncertainty for a body of this scale.

$\text{SNR} = 3$ detection of the baseline tidal signal requires a noise floor of $\sigma_n^* \approx 0.047 \text{ nm s}^{-1}$ (Equation (14)), a factor of 21 improvement over the current SIA specification. Under the source model adopted here ($c = 0.1 \text{ Pa}$), the critical P-wave velocity for tidal detectability with the existing instrument is $V_P^{\text{crit}} \approx 75 \text{ m s}^{-1}$ (Equation (15)), corresponding to shear-wave speeds at the extreme low end of laboratory measurements for granular asteroid analogs. This threshold is model-dependent and would shift upward for stronger source assumptions. A detection of tidal seismicity by SIA would therefore imply an exceptionally unconsolidated interior far below nominal rubble-pile estimates; conversely, a null detection — the most probable outcome — carries scientific value by bounding either V_P or cohesion from below.

Taken together, these results indicate that active seismology is substantially more productive than passive tidal monitoring for the RAMSES mission under current instrument specifications. A 1 kN artificial source — achievable by a spacecraft thruster firing or a small

impactor — produces $\text{SNR} \gg 1$ at all modeled interior properties, enables direct measurement of V_P from P-wave travel times, and provides access to the full normal-mode eigenfrequency spectrum identified here; passive tidal monitoring offers none of these capabilities at the current SIA noise floor. Accordingly, RAMSES should prioritize active-source seismic experiments, and any future SIA instrument upgrade should target $\sigma_n^* \approx 0.047 \text{ nm s}^{-1}$ as the minimum noise floor required to make passive tidal seismometry marginally feasible.

The modeling pipeline introduced here is transferable to any tidally perturbed small body, including Phobos (MMX mission) and binary asteroid systems, establishing shape-accurate spectral-element forward modeling

as a standard pre-mission planning tool for asteroid seismology.

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Facility: JPL Horizons

Software: SPEC-FEM3D v4.1.1 (D. Komatitsch & J. Tromp 1999), NumPy (C. R. Harris et al. 2020), SciPy (P. Virtanen et al. 2020), Matplotlib (J. D. Hunter 2007), Gmsh (C. Geuzaine & J.-F. Remacle 2009)

REFERENCES

- Ballouz, R.-L., Agrusa, H., Barnouin, O., et al. 2024, The Planetary Science Journal, 5, 251, doi: [10.3847/PSJ/ad84f2](https://doi.org/10.3847/PSJ/ad84f2)
- Beaucé, É., Poli, P., Waldhauser, F., Holtzman, B., & Scholz, C. 2023, Geophysical Research Letters, 50, doi: [10.1029/2023gl104375](https://doi.org/10.1029/2023gl104375)
- Binzel, R. P., Rivkin, A. S., Thomas, C. A., et al. 2021, Icarus, 356, 114115, doi: [10.1016/j.icarus.2020.114115](https://doi.org/10.1016/j.icarus.2020.114115)
- Brozovic, M., Benner, L. A. M., Nolan, M. C., et al. 2018, Icarus, 300, 115, doi: [10.1016/j.icarus.2017.08.031](https://doi.org/10.1016/j.icarus.2017.08.031)
- Cadu, A., Garcia, R., Murdoch, N., et al. 2025, in EGU General Assembly Conference Abstracts, doi: [10.5194/egusphere-egu25-17253](https://doi.org/10.5194/egusphere-egu25-17253)
- Cheng, A. F., Agrusa, H. F., Barbee, B. W., et al. 2023, Nature, 616, 457, doi: [10.1038/s41586-023-05810-5](https://doi.org/10.1038/s41586-023-05810-5)
- Daly, R. T., Ernst, C. M., Barnouin, O. S., et al. 2023, Nature, 616, 443, doi: [10.1038/s41586-023-05803-4](https://doi.org/10.1038/s41586-023-05803-4)
- DellaGiustina, D. N., Dworkin, J. P., Nolan, M. C., et al. 2023, Bulletin of the AAS, 55, doi: [10.3847/25c2cfcb.08ccc8c4](https://doi.org/10.3847/25c2cfcb.08ccc8c4)
- DeMartini, J. V., Murdoch, N., Garcia, R. F., & Richardson, D. C. 2024, in EPSC Abstracts, Vol. 18, doi: [10.5194/epsc2024-737](https://doi.org/10.5194/epsc2024-737)
- DeMartini, J. V., Richardson, D. C., Barnouin, O. S., et al. 2019, Icarus, 328, 93, doi: [10.1016/j.icarus.2019.03.015](https://doi.org/10.1016/j.icarus.2019.03.015)
- DeMartini, J. V., Richardson, D. C., Murdoch, N., et al. 2025, The Planetary Science Journal, 6, 263, doi: [10.3847/psj/ae147e](https://doi.org/10.3847/psj/ae147e)
- Desmars, J., Bancelin, D., Hestroffer, D., & Thuillot, W. 2023, Astronomy and Astrophysics, 667, A52, doi: [10.1051/0004-6361/202243880](https://doi.org/10.1051/0004-6361/202243880)
- Dobrovolskis, A. R. 1982, Icarus, 52, 136, doi: [10.1016/0019-1035\(82\)90174-9](https://doi.org/10.1016/0019-1035(82)90174-9)
- Garcia, R. F., Murdoch, N., & Mimoun, D. 2022, Geophysical Research Letters, 49, e2021GL097249, doi: [10.1029/2021GL097249](https://doi.org/10.1029/2021GL097249)
- Geuzaine, C., & Remacle, J.-F. 2009, International Journal for Numerical Methods in Engineering, 79, 1309, doi: [10.1002/nme.2579](https://doi.org/10.1002/nme.2579)
- Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357, doi: [10.1038/s41586-020-2649-2](https://doi.org/10.1038/s41586-020-2649-2)
- Hirabayashi, M., Kim, Y., & Brozovic, M. 2021a, Icarus, 365, 114493, doi: [10.1016/j.icarus.2021.114493](https://doi.org/10.1016/j.icarus.2021.114493)
- Hirabayashi, M., Trowbridge, A. J., Agrusa, H. F., et al. 2021b, Icarus, 365, 114427, doi: [10.1016/j.icarus.2021.114427](https://doi.org/10.1016/j.icarus.2021.114427)
- Hunter, J. D. 2007, Computing in Science & Engineering, 9, 90, doi: [10.1109/MCSE.2007.55](https://doi.org/10.1109/MCSE.2007.55)
- Hurford, T. A., Bills, B. G., Helfenstein, P., et al. 2016, Icarus, 278, 96, doi: [10.1016/j.icarus.2016.06.005](https://doi.org/10.1016/j.icarus.2016.06.005)
- Jin, S., & Ishiguro, M. 2025, Astronomy and Astrophysics, 703, A212, doi: [10.1051/0004-6361/202555854](https://doi.org/10.1051/0004-6361/202555854)
- Kim, Y., Binzel, R. P., Lauretta, D. S., et al. 2023a, The Planetary Science Journal, 4, 214, doi: [10.3847/PSJ/acf7e7](https://doi.org/10.3847/PSJ/acf7e7)

- Kim, Y., DeMartini, J. V., Richardson, D. C., & Hirabayashi, M. 2023b, *Monthly Notices of the Royal Astronomical Society*, 520, 3405, doi: [10.1093/mnras/stad351](https://doi.org/10.1093/mnras/stad351)
- Knapmeyer, M., Murdoch, N., Dehant, V., et al. 2023, *Advances in Space Research*, doi: [10.1016/j.asr.2023.08.036](https://doi.org/10.1016/j.asr.2023.08.036)
- Komatitsch, D., Erlebacher, G., Göddeke, D., & Michéa, D. 2010, *Journal of Computational Physics*, 229, 7692, doi: [10.1016/j.jcp.2010.06.024](https://doi.org/10.1016/j.jcp.2010.06.024)
- Komatitsch, D., & Tromp, J. 1999, *Geophysical Journal International*, 139, 806, doi: [10.1046/j.1365-246x.1999.00967.x](https://doi.org/10.1046/j.1365-246x.1999.00967.x)
- Lazzarin, M., Michel, P., Küppers, M., et al. 2025, in *Proceedings of the International Astronautical Congress (IAC), Space Exploration Symposium*, 405–407, doi: [10.52202/083076-0049](https://doi.org/10.52202/083076-0049)
- Martin, R., Komatitsch, D., Blitz, C., & Le Goff, N. 2008, in *Parallel Computing: Architectures, Algorithms and Applications, Advances in Parallel Computing (IOS Press / Springer)*, 350–363, doi: [10.1007/978-3-540-92859-1_32](https://doi.org/10.1007/978-3-540-92859-1_32)
- Martino, P., Carnelli, I., Küppers, M., Lazzarin, M., & Michel, P. 2024, in *Proceedings of the International Astronautical Congress (IAC), Space Exploration Symposium*, 694–699, doi: [10.52202/078357-0078](https://doi.org/10.52202/078357-0078)
- Murdoch, N., Cadu, A., Garcia, R. F., Sournac, A., & Wilhelm, A. 2025, in *EPSC-DPS Joint Meeting Abstracts*, doi: [10.5194/epsc-dps2025-417](https://doi.org/10.5194/epsc-dps2025-417)
- Murdoch, N., Garcia, R. F., Mimoun, D., et al. 2023, *The Planetary Science Journal*, 4, 94, doi: [10.3847/PSJ/acca23](https://doi.org/10.3847/PSJ/acca23)
- Murdoch, N., Garcia, R. F., Sournac, A., Cadu, A., & Wilhelm, A. 2024, in *EPSC Abstracts*, Vol. 18, doi: [10.5194/epsc2024-495](https://doi.org/10.5194/epsc2024-495)
- Murdoch, N., Hempel, S., Pou, L., et al. 2015a, *Journal of Geophysical Research: Planets*, 120, 1767, doi: [10.1002/2015JE004810](https://doi.org/10.1002/2015JE004810)
- Murdoch, N., Sanchez, P., Schwartz, S. R., & Miyamoto, H. 2015b, in *Asteroids IV*, ed. P. Michel, F. E. DeMeo, & W. F. Bottke (University of Arizona Press), doi: [10.2458/azu_uapress.9780816532131-ch027](https://doi.org/10.2458/azu_uapress.9780816532131-ch027)
- Naidu, S. P., Benner, L. A. M., Brozovic, M., et al. 2016, *The Astronomical Journal*, 152, 99, doi: [10.3847/0004-6256/152/4/99](https://doi.org/10.3847/0004-6256/152/4/99)
- Naidu, S. P., Benner, L. A. M., Brozovic, M., et al. 2022, *Icarus*, 348, 113820, doi: [10.1016/j.icarus.2020.113820](https://doi.org/10.1016/j.icarus.2020.113820)
- Nakamura, Y., Latham, G., Dorman, J. H., et al. 1979, *Proceedings of the Lunar and Planetary Science Conference*, 10, 2299, doi: [10.1029/LB-i10p2299](https://doi.org/10.1029/LB-i10p2299)
- Nimmo, F., & Matsuyama, I. 2019, *Icarus*, 321, 715, doi: [10.1016/j.icarus.2018.09.007](https://doi.org/10.1016/j.icarus.2018.09.007)
- Pou, L., Murdoch, N., Mimoun, D., Garcia, R. F., & Margerin, L. 2020, *Geophysical Research Letters*, 47, e2019GL085603, doi: [10.1029/2019GL085603](https://doi.org/10.1029/2019GL085603)
- Quillen, A. C., Giannella, D., Shaw, J. G., & Ebel, C. 2022, *Icarus*, 380, 114962, doi: [10.1016/j.icarus.2022.114962](https://doi.org/10.1016/j.icarus.2022.114962)
- Raducan, S. D., Jutzi, M., Cheng, A. F., et al. 2024, *Nature Astronomy*, 8, 445, doi: [10.1038/s41550-024-02200-3](https://doi.org/10.1038/s41550-024-02200-3)
- Sánchez, P., & Scheeres, D. J. 2014, *Meteoritics and Planetary Science*, 49, 788, doi: [10.1111/maps.12293](https://doi.org/10.1111/maps.12293)
- Virtanen, P., Gommers, R., Oliphant, T. E., et al. 2020, *Nature Methods*, 17, 261, doi: [10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2)
- Wang, W., Shearer, P., Vidale, J., et al. 2022, *Earth and Planetary Science Letters*, 579, 117335, doi: [10.1016/j.epsl.2021.117335](https://doi.org/10.1016/j.epsl.2021.117335)
- Zhang, Y., Michel, P., Richardson, D. C., et al. 2021, *Icarus*, 362, 114433, doi: [10.1016/j.icarus.2021.114433](https://doi.org/10.1016/j.icarus.2021.114433)